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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Code Queens
Selected Domain	Domain 1 — Aiversecity Future-Ready Workforce
Selected SDG Track (1–17)	SDG 4 — Quality Education; Target 4.4
Project / Solution Name	NextArc
Name (Leader)	Chew Khai Shuen
Email & Phone (Leader)	23034093@imail.sunway.edu.my (012-4085129)
University Affiliation(s)	Sunway University
Number of Team Members	4

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Fresh graduates need to decide which skills to learn as AI tools change the tasks associated with different careers. Graduates with a preferred direction need relevant learning priorities; those who are undecided need a way to explore options while retaining control of their choice.

Our problem hypothesis is that graduates struggle to connect information about AI developments with their existing skills, a career route and an achievable learning plan. Choosing unsuitable resources may consume limited time and money without addressing the skills needed for their intended work.

The World Economic Forum's Future of Jobs Report 2025 estimates that 39% of workers' existing skill sets will change or become outdated during 2025–2030. This is a global employer outlook, not a measured result for our graduate audience [1].

Our intended audience is fresh graduates. For preliminary testing and validation, we focus on CS fresh graduates considering Backend Development and QA Automation. This limited pilot allows us to check capability mappings, explanations and learning resources consistently before considering expansion.

We aim to support relevant technical learning, affordable resources and informed career decisions. Our exploratory manual-search pilot provides an initial, limited baseline [9]; broader graduate needs and the usefulness of our recommendations still require evaluation.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

NextArc helps fresh graduates choose what to learn next and understand why. Its pretrained Gemini adapter can extract draft skills from fictional CVs. Users correct errors, add omitted skills and confirm their profile before matching. Graduates can keep a preferred backend or QA route, or compare both before choosing.

Role scores show the percentage of listed capabilities the graduate has confirmed, rounded to a whole number. Each capability counts once; unrelated skills incur no penalty and proficiency adds no bonus. Missing or uncertain skills receive no credit yet, but users still receive results and can update their answers. A score describes our pilot capability list, not employability or hiring probability.

The deployed app uses a sourced baseline with 16 capabilities and 18 documentation/occupational evidence records. Versioned criteria connect role comparison, learning pathways, supporting sources and SDG explanations. Pathways place prerequisites before dependent skills and label estimated practice time.

We have implemented bounded GitHub Changelog and JetBrains AI retrieval, Gemini interpretation and fixed admission checks. A tool name alone cannot create a scored capability. The public release currently has trend interpretation disabled; a successful live source-to-advice update has not been demonstrated.

For the final product, the updateable trend layer will support evidence-based learning advice, with publication dates, check status and source limitations. Each role and recommended skill connects to SDG 4, Target 4.4. We also plan moderate and faster AI-adoption scenarios with visible assumptions and estimated learning implications. Tool announcements will not be presented as proof of hiring growth or decline.

This covers delivery of all six Domain 1 features [3], combining GenAI APIs, database design and Replit hosting without training or fine-tuning a model.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: plain HTML, CSS and JavaScript ES modules provide CV upload, profile review, role comparison and a metro-style learning roadmap. Supported uploads are text-based PDFs or UTF-8 TXT files up to 2 MB. The hackathon prototype uses fictional CVs only; scanned-image OCR is not implemented.

Backend and database: Node.js 24 runs TypeScript directly, with Express, Zod validation, Multer and unpdf. Prisma and the PostgreSQL driver connect to Neon. Uploaded CVs and personal profile state are processed in memory rather than saved to the application database. Public criteria, evidence, source provenance and refresh state use versioned database storage.

AI and sources: @google/genai supports Gemini CV extraction and a separate bounded trend interpreter. LLM_PROVIDER and GEMINI_MODEL configure extraction; TREND_PROVIDER and TREND_GEMINI_MODEL configure trend interpretation. The public trend setting is currently disabled. Exact model identities and service settings will be disclosed in the Technical & Tools Declaration. O*NET 31.0 supplies occupational context, not Malaysian graduate vacancy statistics [5]. GitHub Changelog and JetBrains AI are the configured update feeds [6–7]. fast-xml-parser parses and validates RSS.

User flow: browser → CV extraction → user correction and confirmation → version-pinned role comparison → chosen-route pathway, resources and SDG explanations.

Update flow, when enabled: publisher feeds → bounded Gemini interpretation → admission and deduplication checks → saved criteria version → the same matching and pathway readers. Failed updates retain the last valid criteria. Confirmation or an explicit user update check can trigger a bounded refresh; there is no continuous polling. Shared cooldown and in-flight controls limit duplicate calls. Automated checks constrain model output but do not prove semantic correctness or occupational necessity.

Hosting and sponsor platform: Replit is used for development and public Autoscale hosting. GitHub stores source, dependency locks and incremental commits. No sponsor credit amount or RunPod/GPU execution is claimed here.

Feature 2 is integrated into main at 0aebfdb, including the criteria bootstrap command [10]. The public app serves the new frontend/API, two eight-capability role lists, 16 recognised capabilities and 18 baseline evidence records [11]. Live AI-trend refresh is implemented but currently unavailable in the public release due to limitations under our subscription plan. The app therefore provides

recommendations using the sourced baseline. Latest recorded checks passed 210 backend tests, 60 frontend tests and TypeScript checking. Nine PostgreSQL checks, live-model validation and complete deployed-journey verification remain outstanding.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We narrowed a broad graduate-learning problem to two CS career routes so four members can build and evaluate a coherent prototype. Published workforce context informed the problem. Our exploratory manual-search pilot provides initial evidence of difficulty selecting relevant skills, but its small, mixed sample does not establish the needs of all fresh graduates [9].

Two fictional personas cover backend and QA backgrounds, with preferred-route and exploration journeys. Users review AI extraction and retain the final career choice. The design avoids treating a CV omission as proof of inability and explains the limited meaning of a coverage score.

We reused the existing application and integrated the frontend with versioned criteria. Fictional browser checks covered profile corrections, route choice, skill-answer updates, prerequisites, source status and failure cases. The compatible frontend/API and sourced baseline are now served on Replit. Live trend interpretation remains disabled pending verification.

Next, we will verify real-model behaviour, remaining PostgreSQL checks and both complete deployed journeys. We will compare manual and app-assisted skill selection using consistent tasks, recording completion, perceived relevance and time. Final-round work includes verified live updates, multiple estimated scenarios and broader evaluation before route expansion.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 4 — Quality Education, particularly Target 4.4: increasing relevant technical and vocational skills for employment, decent jobs and entrepreneurship [2]. NextArc supports this target by helping graduates identify relevant learning priorities and access free or low-cost resources. Its immediate contribution is learning guidance; selecting a pathway does not itself demonstrate skill acquisition or improved employment outcomes.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	An exploratory manual-search pilot involved 10 participants, including teammates, undergraduates seeking internships and fresh graduates. Participants searched for recent AI-related skills they considered relevant to their intended career routes. Five participants selected at least one relevant skill, giving an initial self-reported skill-selection completion rate of 50%. Observed search times ranged from approximately 10–30 minutes. This small, mixed sample provides an initial pilot baseline rather than a representative finding about fresh graduates. Participants judged relevance themselves; the pilot did not independently verify recommendation quality, résumé improvement or skill acquisition. Source: Code Queens team exploratory pilot, 10 participants.
Core Measurable KPI *The specific unit you will track*	**Skill-selection completion rate:** the percentage of participants who select at least one recent AI-related skill they consider

Metric Category	Your Response
e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.	relevant to their intended career role and potentially useful for strengthening their résumé. Calculate: participants selecting a skill ÷ total participants × 100. The initial manual-search baseline was **5 out of 10 participants (50%)**. Record search time separately using consistent start and end points.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	By the Phase 2 prototype presentation, we aim for at least 80% of participants to select an AI-related skill they consider relevant to their intended career role, compared with the initial manual-search baseline of 50%. We also target a 40% reduction in median skill-search time. Recommendations will include dated sources and clear explanations connecting skills to career roles, helping users make better-informed learning decisions.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

Draft note: workshop completion, final evidence and all-fields readiness need leader verification. Checkboxes are intentionally left for the team; the proposal is not yet submission-ready.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes √ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

[Leader and members to confirm Yes/No and disclose any pre-hackathon work; no answer is inferred.]

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Express 5.3.0; Zod 4.6.7; fast-xml-parser 5.11.2; Multer 2.4.0; pg 8.23.1; unpdf 1.9.0; MIT-licensed @types declarations	Code libraries	MIT; retain each package's notice
TypeScript 7.0.2; Prisma client, adapter and CLI 7.10.1; @google/genai 2.30.0	Code libraries and development tools	Apache-2.0; retain applicable notices
PDF.js 6.4.299 bundled with unpdf	PDF parsing library	Apache-2.0; retain bundled attribution and notices
`@types/express` 5.0.6; `@types/node` 26.6.5; `@types/multer` 2.3.0; `@types/pg` 8.23.1	Development type declarations	MIT
Node.js 24; PostgreSQL;	Runtime and database	Node.js MIT and bundled third-party notices; PostgreSQL License

Name	Type (code/dataset/model)	License
O*NET 31.0, filtered to Software Developers and Software Quality Assurance Analysts and Testers	Public occupational dataset	CC BY 4.0; USDOL/ETA attribution; filtered and adapted [5]
Public technical documentation and learning-resource metadata	Application reference content	Publisher/provider terms; linked content retains its original rights.
GitHub Changelog; JetBrains AI; linked learning documentation	Public AI-update sources for the implemented trend pipeline	Publisher copyright and applicable terms; links and brief interpretations [6–7].
Google Gemini through the API	Pretrained model integration for CV extraction and AI-trend interpretation	Google API and Gemini API service terms; exact configured model IDs belong in the Technical & Tools Declaration.
IBM Plex Sans, Sans Condensed and Mono, delivered through Google Fonts	Frontend font assets	SIL Open Font License 1.1.
Replit; Neon; Github	Application hosting, managed PostgreSQL and source hosting	Respective provider service terms; separate from the project's source-code licence.

Team-authored project code is licensed under Apache-2.0. Third-party packages, datasets, fonts and provider content retain their own terms. O*NET mappings are filtered and adapted by Code Queens without USDOL/ETA endorsement. The application integrates pretrained models without custom training or fine-tuning. Development assistance from Replit Agent, Claude Code and Codex, including implementation, testing, review and documentation, is disclosed in the separate Technical & Tools Declaration.

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes √ No

If yes, please describe the nature of the agreement and which member it concerns:

[Leader and members to confirm employment/internship/consulting IP obligations before signing.]

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Khai Shuen

Chew Khai Shuen

[Signature]

[Date] 25/08/2026

Appendix A Sources and draft status

[1] World Economic Forum, “The Future of Jobs Report 2025,” Jan. 7, 2025. [Online]. Available: https://www.weforum.org/publications/the-future-of-jobs-report-2025

[2] United Nations. Goal 4, Target 4.4 and indicator 4.4.1.

https://sdgs.un.org/goals/goal4

[3] Aiversecity Domain 1. Sections 3–8: journey, six must-have features, boundaries and preliminary rubric.

https://drive.google.com/file/d/1x6Nvjl_1Gc-v3SsTnf44HVEJZQsbWVRA/view

[4] SDC 2026 official Preliminary Project Proposal template. This editable draft retains that form.

https://docs.google.com/document/d/1RUb7bQvtaI5ds8SLHQtzAFmg8OGb9WsK/edit

[5] O*NET 31.0 Database, U.S. Department of Labor, Employment and Training Administration. Filtered to software developer and QA occupations; app task/skill mappings are team adaptations without USDOL/ETA endorsement. CC BY 4.0.

https://www.onetcenter.org/database.html | https://www.onetcenter.org/license_db.html

[6] GitHub Changelog public feed. Evidence of product capabilities, not hiring demand.

https://github.blog/changelog/feed/

[7] JetBrains AI public feed. Evidence of product capabilities, not hiring demand.

https://blog.jetbrains.com/ai/feed/

[8] Participant Handbook. PRD requirement and submission instructions; use the live portal for final submission.

https://drive.google.com/file/d/1fGBS-jevDj_FrF8wsbYNPzyU5TTrfk1V/view

Draft status: name selected by Jong; numerical task baseline and 20% impact target are proposed for leader review. Implementation checkpoint: f578bd9 on codex/feature-2-automation. Frontend integration, real-model/database checks and release evidence remain pending in this checkpoint; replace with verified team facts before submission.